

Nurture Course COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2023

Test Type : PRACTICE TEST

Test Pattern : NEET (UG)

TEST DATE : _____

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	1	2	2	2	3	1	2	1	3	1	4	2	3	3	3	4	3	1	4	4	2	2	3	1	4	1	3	4	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	1	4	1	2	2	2	1	2	3	1	3	1	3	4	4															



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HINT - SHEET

1. **Ans (1)**

Using slope of the curve we can write equation

$$a = -10x$$

Comparing it with $a = -\omega^2 x$

$$\omega^2 = 10$$

$$\omega = \sqrt{10}$$

$$\text{frequency} = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{10}$$

2. **Ans (1)**

Use general equation of SHM

$$y = a \sin(\omega t + \phi)$$

$$= 0.8 \sin \left\{ \frac{2\pi t}{0.2} + \frac{\pi}{2} \right\}$$

$$= 0.8 \sin \left\{ 10\pi t + \frac{\pi}{2} \right\}$$

$$= 0.8 \cos 10\pi t$$

3. **Ans (2)**

$$\text{Maximum Kinetic Energy} = \frac{1}{2} m \omega^2 A^2$$

$$\omega = \sqrt{\frac{g}{L}}, A = L\theta$$

$$\text{KE} = \frac{1}{2} m \frac{g}{L} \times L^2 \theta^2, \text{KE} = \frac{1}{2} mgL\theta^2$$

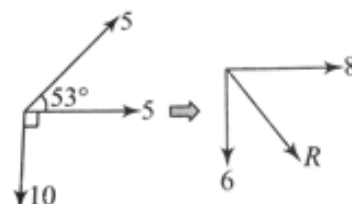
$$\Rightarrow K_1 = \frac{1}{2} mgL\theta^2$$

If Length is double, [Here we are assuming angular amplitude is same]

$$K_2 = \frac{1}{2} mg(2L)\theta^2$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{\frac{1}{2} mgL\theta^2}{\frac{1}{2} mg(2L)\theta^2} = \frac{1}{2}$$

4. **Ans (2)**



For equation no - 3

$$X_3 = 10 \sin(\omega t + 270^\circ)$$

From the Phasor diagram, By using principle of superposition resultant amplitude is given by

$$R = \sqrt{8^2 + 6^2} = 10$$

5. **Ans (2)**

From spring block system,

Time period is given by - $T = 2\pi\sqrt{\frac{m}{k}}$

Here, $t_1 = 2\pi\sqrt{\frac{m}{k_1}}$ and $t_2 = 2\pi\sqrt{\frac{m}{k_2}}$

Now, for series combination of spring,

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$$

$$\begin{aligned} T^2 &= 4\pi^2 \frac{m}{k} \\ &= 4\pi^2 \left(\frac{m}{k_1} + \frac{m}{k_2} \right) \\ &= t_1^2 + t_2^2 \end{aligned}$$

6. **Ans (3)**

When $t = \frac{T}{12}$

$$\text{Then } x = A \sin \left(\frac{2\pi}{T} \times \frac{T}{12} \right) = \frac{A}{2}$$

$$\begin{aligned} KE &= \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 (A^2 - x^2) \\ &= \frac{1}{2}m\omega^2 \left(A^2 - \frac{A^2}{4} \right) = \frac{3}{4} \left(\frac{1}{2}m\omega^2 A^2 \right) \end{aligned}$$

$$PE = \frac{1}{2}m\omega^2 x^2 = \frac{1}{4} \left(\frac{1}{2}m\omega^2 A^2 \right)$$

$$\frac{KE}{PE} = \frac{3}{1}$$

7. **Ans (1)**

$$\begin{aligned} T &= 2\pi\sqrt{\frac{L}{g}} \\ &= 2\pi\sqrt{\frac{L(R+h)^2}{GM}} \end{aligned}$$

$$T \propto (R+h)$$

$$\text{or } \frac{T'}{T} = \frac{R+h}{R}$$

$$= 1 + \frac{h}{R} = 1 + \frac{1}{100}$$

$$\text{or } \frac{T' - T}{T} \times 100 = 1$$

Time period will increase by 1%

8. **Ans (2)**

If m is the mass and r is the amplitude of oscillation, then maximum kinetic energy,

$$k_0 = \frac{1}{2}m\omega r$$

$$r = \left(\frac{2k_0}{m\omega} \right)^{\frac{1}{2}}$$

The displacement equation can be

$$y = r \sin \omega t = \left(\frac{2k_0}{m\omega} \right)^{\frac{1}{2}} \sin \omega t$$

9. **Ans (1)**

$$y = A \sin \omega t + B$$

The amplitude of the simple harmonic motion is A .

10. **Ans (3)**

In the diagram B denotes the mean position.

$$BC = AB \sin \left(\frac{2\pi}{T} t_{BC} \right)$$

Let, T be the time period of oscillation of the spring mass system, Time from B to C , t_{BC} can be given by

$$\frac{BC}{AB} = \frac{x_0}{2x_0} t_{BC} = \frac{1}{2}$$

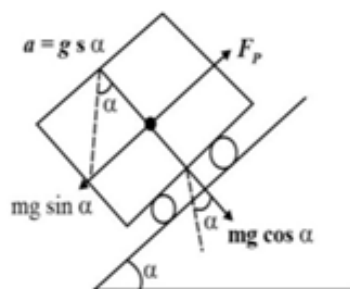
$$\Rightarrow t_{BC} = \frac{T}{12}$$

Total time from A to C

$$t_{AC} = t_{AB} + t_{BC} = \frac{T}{4} + \frac{T}{12} = \frac{T}{3}$$

$$\Rightarrow t_{AC} = \frac{2\pi}{3} \sqrt{\frac{m}{k}}$$

11. **Ans (1)**



When vehicle is accelerated F_{pseudo} will act on pendulum which will balance the force along the incline

$$F_p = mg \sin \alpha$$

$$\text{Therefore, } g_{\text{eff}} = g \cos \alpha$$

So,

$$T = 2\pi\sqrt{\frac{L}{g \cos \alpha}}$$

12. **Ans (4)**

The total energy E of a particle vibrating in SHM is given by

$$E = \frac{1}{2} m \omega^2 a^2 \dots (1)$$

The kinetic energy K is given by

$$K = \frac{1}{2} m \omega^2 (a^2 - y^2)$$

Where y = displacement of the particle, but

$$K = \frac{E}{2} = \frac{1}{2} \left[\frac{1}{2} m \omega^2 a^2 \right]$$

$$\Rightarrow \therefore \frac{1}{2} \left[\frac{1}{2} m \omega^2 a^2 \right] = \frac{1}{2} m \omega^2 (a^2 - y^2)$$

$$\text{or } \frac{a^2}{2} = a^2 - y^2$$

$$\text{or } y^2 = \frac{a^2}{2}$$

$$\Rightarrow \therefore y = \frac{a}{\sqrt{2}}$$

$$\Rightarrow \therefore y = \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ cm}$$

13. **Ans (2)**

Acceleration $|a| = \omega^2 x$

$$\text{or } \omega = \sqrt{\frac{a}{x}} = \sqrt{\frac{2}{0.02}} = 10 \text{ rad s}^{-1}$$

14. **Ans (3)**

Initially, $T = 2\pi \sqrt{\frac{M}{k}}$ and

$$\text{Finally, } \frac{5T}{3} = 2\pi \sqrt{\frac{M+m}{k}}$$

$$\therefore \frac{5}{3} \times 2\pi \sqrt{\frac{M}{k}} = 2\pi \sqrt{\frac{M+m}{k}}$$

$$\text{or } \frac{25}{9} \frac{M}{k} = \frac{M+m}{k}$$

$$\text{or } 9m + 9M = 25M$$

$$\text{or } \frac{m}{M} = \frac{16}{9}$$

15. **Ans (3)**

$$V_1 = \frac{d}{dt}(y_1) = (0.1 \times 100\pi) \cos(100\pi t + \frac{\pi}{3})$$

$$V_2 = \frac{d}{dt}(y_2) = (-0.1 \times \pi) \sin \pi t$$

$$= (0.1 \times \pi) \cos(\pi t + \frac{\pi}{2})$$

$$\therefore \Delta \phi = \frac{\pi}{3} - \frac{\pi}{2} = -\frac{\pi}{6}$$

16. **Ans (3)**

$$m\omega^2 a = mg, \omega = \sqrt{\frac{g}{a}}, T = 2\pi \sqrt{\frac{a}{g}},$$

$$T = 2\pi \sqrt{\frac{2.5}{1000}} = \frac{\pi}{10} \text{ s}$$

17. **Ans (4)**

$$x = x_0 \cos^2 \omega t$$

$$x = x_0 \left(\frac{\cos 2\omega t + 1}{2} \right)$$

$$x = \frac{x_0}{2} + \frac{x_0}{2} \sin \omega t \cdot 2\omega$$

$$\frac{dx}{dt} = -\frac{x_0}{2} \sin 2\omega t \cdot 2\omega$$

$$\frac{dx}{dt} = -x_0 \omega \sin 2\omega t$$

So the speed at its mean position is $x_0 \omega$.

18. **Ans (3)**

$$y = 0.2 \sin(10\pi t + 1.5\pi) \cos(10\pi t + 1.5\pi)$$

$$= 0.1 \sin 2(10\pi t + 1.5\pi) (\sin 2A = 2 \sin A \cos A)$$

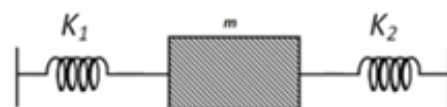
$$= 0.1 \sin(20\pi t + 3.0\pi)$$

$$\therefore \text{Time period } T = \frac{2\pi}{\omega} = \frac{2\pi}{20\pi} = \frac{1}{10} = 0.1 \text{ s}$$

19. **Ans (1)**

As $T_1 > T_2$; Hence Time period first increases and then decreases to the original value

20. **Ans (4)**



In the given figure, two springs are connected in parallel. Therefore, the effective spring constant is given by

$$K_{\text{eff}} = K_1 + K_2$$

Frequency of oscillation,

$$f = \frac{1}{2\pi} \sqrt{\frac{K_{\text{eff}}}{m}} = \frac{1}{2\pi} \sqrt{\frac{K_1 + K_2}{m}}$$

As K_1 and K_2 are increased by four times.

New frequency

$$f' = \frac{1}{2\pi} \sqrt{\frac{4(K_1 + K_2)}{m}} = 2f$$

21. **Ans (4)**

In case of cutting of a spring

$$l \times k = \text{constant}$$

$$k_1 l_1 = k_2 l_2$$

$$k_1 (\eta l_2) = k_2 l_2$$

$$\Rightarrow \eta k_1 = k_2 \quad \dots (1)$$

Now both the spring s when connected in series should give the same effect as that of the original spring. Hence, by applying the concept of springs connected in series, we obtain,

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \quad \dots \dots (2)$$

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{\eta k_1}$$

$$\frac{k_1}{k} = \frac{\eta + 1}{\eta} \quad (\text{using (1)})$$

$$k_1 = \left(\frac{\eta + 1}{\eta} \right) k \quad \dots \dots \dots (3)$$

Using (3) again in (2), we obtained,

$$k_2 = (\eta + 1)k$$

22. **Ans (2)**

$$P.E. = \frac{1}{2} m \omega^2 x^2 = 2.5J$$

$$\text{As } x = \frac{A}{2}$$

$$\frac{1}{2} m \omega^2 \left(\frac{A}{2} \right)^2 = 2.5$$

$$\therefore \frac{1}{2} m \omega^2 \frac{A^2}{4} = 2.5$$

$$\therefore \frac{1}{2} m \omega^2 A^2 = 10J$$

Total energy of system

$$= \frac{1}{2} m \omega^2 A^2 = 10J$$

23. **Ans (2)**

Energy of a damping oscillator is

$$E = E_0 e^{-\gamma t} \text{ at time } t$$

(where γ = damping constant)

$$E' = E_0 e^{-\gamma t}$$

$$15 = 45 e^{-\gamma(15)}$$

$$\gamma = \frac{1}{15} \ln 3$$

24. **Ans (3)**

The motion of block is S.H.M.

$$Y = A \sin (\omega t + \phi)$$

Here amplitude is

$$A = \frac{mg}{k} = \frac{20 \times 10}{2000} m = 10cm$$

At $t = 0$, displacement of body with respect to mean position is, $Y = 15 - 10 = 5 \text{ cm}$

$$\therefore 5 = 10 \sin (\omega \times 0 + \phi)$$

$$\text{or } \frac{1}{2} = \sin \phi \Rightarrow \phi = \frac{\pi}{6}$$

$$\therefore y = 10 \sin \left(10t + \frac{\pi}{6} \right)$$

25. **Ans (1)**

$$x = A \sin \omega t$$

$$\text{For } x = \frac{A}{2}, \sin \omega T_1 = \frac{1}{2} \Rightarrow T_1 = \frac{\pi}{6\omega}$$

$$\text{For } x = A, \sin \omega (T_1 + T_2) = 1$$

$$\Rightarrow T_1 + T_2 = \frac{\pi}{2\omega}$$

$$\Rightarrow T_2 = \frac{\pi}{2\omega} - T_1 = \frac{\pi}{2\omega} - \frac{\pi}{6\omega} = \frac{\pi}{3\omega}$$

$$\text{i.e, } T_1 < T_2$$

26. **Ans (4)**

The time period of a simple pendulum length l , is given by

$$T = 2\pi \sqrt{\frac{l}{g}} = \frac{1}{\text{frequency (n)}}$$

Where g is the acceleration due to gravity

$$\frac{n_1}{n_2} = \sqrt{\frac{l_2}{l_1}}$$

$$\frac{l_1}{l_2} = \left(\frac{n_2}{n_1} \right)^2$$

$$\frac{n_2}{n_1} = \frac{3}{2}$$

$$\frac{l_1}{l_2} = \left(\frac{n_2}{n_1} \right)^2 = \left(\frac{3}{2} \right)^2 = \frac{9}{4}$$

27. **Ans (1)**

As force is always in the direction of acceleration.

28. **Ans (3)**

$$10 = \omega\sqrt{a-16} \quad \dots(1)$$

$$8 = \omega\sqrt{a-25} \quad \dots(2)$$

$$\left(\frac{5}{4}\right)^2 = \frac{a^2-16}{a^2-25}$$

$$\text{or } \frac{25}{16} = \frac{a^2-16}{a^2-25}$$

$$\Rightarrow 9a^2 = 625 - 256$$

$$\Rightarrow a = \frac{369}{9} = 41$$

From equation (i)

$$10 = \omega\sqrt{41-16} = \omega \times 5$$

$$\text{or } \omega = \frac{2\pi}{T} = 2$$

$$\Rightarrow T = \pi \text{ s}$$

29. **Ans (4)**

$$E_1 = E_2$$

$$\therefore \frac{1}{2}m_1\omega_1^2A_1^2 = \frac{1}{2}m_2\omega_2^2A_2^2$$

$$\text{but } m_1 = m_2$$

$$\therefore \omega_1^2 \times 16 = \omega_2^2 \times 25$$

$$\therefore 100 \times 16 = \omega^2 \times 25$$

$$\omega = 8 \text{ units}$$

30. **Ans (2)**

Using the general equation of SHM

$$x = A \sin \omega t$$

as per graph $A = 1$, $T = 8 \text{ sec}$

$$\omega = \frac{2\pi}{T} = \frac{\pi}{4} \Rightarrow x = \sin \frac{\pi}{4} t$$

$$\text{acceleration } a = \frac{d^2x}{dt^2}$$

$$\Rightarrow a = -\frac{\pi^2}{16} \sin \frac{\pi}{4} t$$

$$\text{at } t = \frac{2}{3}$$

$$a = -\frac{\pi^2}{16} \sin\left(\frac{\pi}{4} \cdot \frac{2}{3}\right)$$

$$= -\frac{\pi^2}{16} \sin\left(\frac{\pi}{6}\right)$$

$$\Rightarrow a = -\frac{\pi^2}{16} \times \frac{1}{2} = -\frac{\pi^2}{32} \frac{\text{cm}}{\text{s}^2}$$

31. **Ans (1)**

Comparing $a + 4\pi^2x = 0$ with standard equation of simple harmonic motion

$$a + \omega^2x = 0$$

$$\omega^2 = 4\pi^2, \omega = 2\pi$$

$$2\pi n = 2\pi, n = 1 \text{ Hz}$$

32. **Ans (4)**

$$T = 8 \text{ s}$$

$$\omega = \frac{2\pi}{T} = \frac{\pi}{4}$$

$$\text{Equation of SHM } y = A \sin \frac{\pi}{4} t$$

$$y(1) = A \sin\left(\frac{\pi}{4}\right)$$

$$y(2) = A \sin\left(\frac{\pi}{2}\right) = A$$

Distance travelled in 2nd second

$$y(2) - y(1) = A - \frac{A}{\sqrt{2}}$$

$$\text{So the ratio would be } = \frac{\frac{A}{\sqrt{2}}}{\left(A - \frac{A}{\sqrt{2}}\right)}$$

$$= \frac{1}{(\sqrt{2} - 1)}$$

$$= \frac{[(\sqrt{2} + 1)]}{(\sqrt{2} - 1)(\sqrt{2} + 1)} = \sqrt{2} + 1$$

33. **Ans (1)**

In first case, springs are connected in parallel, so their equivalent spring constant

$$k_p = k_1 + k_2$$

So, frequency of this spring-block system is

$$f_p = \frac{1}{2\pi} \sqrt{\frac{k_p}{m}}$$

$$\text{or } f_p = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}}$$

$$\text{but } k_1 = k_2 = k$$

$$\therefore f_p = \frac{1}{2\pi} \sqrt{\frac{2k}{m}} \dots \dots (i)$$

Now in second case, springs are connected in series, so equivalent spring constant

$$k = \frac{k_1 k_2}{k_1 + k_2}$$

Hence frequency of this arrangement is given by

$$f_s = \frac{1}{2\pi} \sqrt{\frac{k_1 k_2}{(k_1 + k_2) m}}$$

$$\text{or } f_s = \frac{1}{2\pi} \sqrt{\frac{k}{2m}} \dots (ii)$$

Dividing eq (ii) by (i), we get

$$\frac{f_s}{f_p} = \frac{\frac{1}{2\pi} \sqrt{\frac{k}{2m}}}{\frac{1}{2\pi} \sqrt{\frac{2k}{m}}} = \sqrt{\frac{1}{4}}$$

$$\text{or } \frac{f_s}{f_p} = \frac{1}{2}$$

34. **Ans (2)**

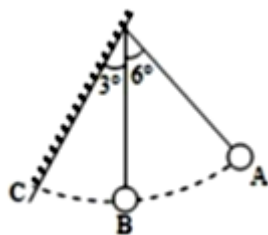
$$a = -\beta(x - 2)$$

$$\Rightarrow a = -\omega^2(x - x_0)$$

By comparing

$$\omega^2 = \beta \Rightarrow T = 2\pi\sqrt{\frac{1}{\beta}}$$

35. **Ans (2)**



$$\text{Let } T = 2\pi\sqrt{\frac{1}{g}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{0.1}{10}} = \frac{\pi}{5} \text{ s}$$

Time taken by the pendulum in going from A

$$\text{to B} = \frac{T}{4}$$

Time taken by the pendulum in going from B to

$$C = \frac{T}{12}$$

Time period of the pendulum,

$$= 2 \left(\frac{T}{4} + \frac{T}{12} \right) = \frac{2T}{3} = \frac{2}{3} \cdot \frac{\pi}{5} = \frac{2\pi}{15} \text{ s}$$

36. **Ans (2)**

Let displacement be

$$x = A \sin(\omega t + \phi)$$

Kinetic energy,

$$K = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t + \phi)$$

As the average value of $\cos^2(\omega t + \phi)$ over a

cycle is $\frac{1}{2}$.

Therefore average kinetic energy is

$$K_{av} = \frac{1}{4} m \omega^2 A^2$$

37. **Ans (1)**

It is given that $F = -k\sqrt{x}$

But For SHM $F = -k_0x$

Therefore by comparing them $k = -k_0\sqrt{x}$

Therefore as x increases k increases.

38. **Ans (2)**

$$T \propto \frac{1}{\sqrt{K}} \text{ and } K \propto \frac{1}{l}$$

$$\therefore T \propto \sqrt{l}$$

Hence the answer is $1 : \sqrt{3} : \sqrt{5}$

39. **Ans (3)**

$$a = \frac{t^2}{m} - \frac{k}{m}x$$

$$\frac{da}{dt} = \frac{1}{m}(2t) - \frac{k}{m} \cdot \frac{dx}{dt}$$

$$\frac{d^2a}{dt^2} = \frac{2}{m} - \frac{k}{m}a$$

$$\frac{d^2a}{dt^2} + \frac{k}{m}a - \frac{2}{m} = 0$$

This is second order differential equation for acceleration, hence a will be in simple harmonic

40. **Ans (1)**

When m_1 is removed, only m_2 is left.

Therefore, angular frequency

$$\omega = \sqrt{\frac{k}{m_2}}$$

41. **Ans (3)**

When lift accelerates upwards, then effective acceleration on the pendulum

$$g_{\text{eff}} = g + \frac{g}{3} = \frac{4g}{3}$$

$$\therefore \text{Time period } T' = 2\pi\sqrt{\frac{1}{g_{\text{eff}}}}$$

$$= 2\pi\sqrt{\frac{1}{\frac{4g}{3}}} = \frac{\sqrt{3}}{2} \cdot 2\pi\sqrt{\frac{1}{g}} = \frac{\sqrt{3}}{2} T$$

42. **Ans (1)**

$$\text{Potential Energy} = \frac{1}{2} M \omega^2 A^2$$

$$= \frac{1}{2} M \cdot \frac{g}{L} \cdot A^2 \left(\because \omega = \sqrt{\frac{g}{L}} \right)$$

43. **Ans (3)**

δ is the initial phase. When oscillations of pendulum are damped, δ doesn't change

44. **Ans (4)**

Amplitude $A = 0.06$ m

Frequency, $f = 15$ Hz

Maximum velocity of the body,

$$v_m = \omega A = (2\pi f) A$$

$$= 2 \times 3.14 \times 15 \times 0.06 = 5.65 \text{ m/s}$$

Maximum acceleration of the body

$$a_m = \omega^2 A = (2\pi f)^2 A = 4\pi^2 f^2 A$$

$$= 4 \times 3.14 \times 15 \times 0.06$$

$$= 5.32 \times 10^2 \text{ m/s}^2$$

45. **Ans (4)**

Given :

$$\text{displacement } x = 2 \times 10^{-2} \cos \pi t$$

$$\text{Velocity } v = \frac{dx}{dt} = -2 \times 10^{-2} \pi \sin \pi t$$

For the first time when

$$v = v_{\max}, \sin \pi t = 1$$

$$\sin \pi t = \sin \frac{\pi}{2}$$

$$\text{or } \pi t = \frac{\pi}{2}$$

$$t = \frac{1}{2} \text{ s} = 0.5 \text{ s}$$